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TRW Automotive

Slip Control Software Design

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1 ABSTRACT

This module provides common functions for Active Roll Mitigation ([ARM](#)) algorithm such as filters.

2 ACRONYMS

ARM Active Roll Mitigation

AyMod 'Modified' Lateral Acceleration, that tries to predict the Lateral Acceleration

3 OVERVIEW

This document describes the purpose of Arm.Utilities.

3.1 DESIGN GOAL

Some common functions are provided for example filter, with special functionality

3.2 GENERAL REQUIREMENTS

1. [ARM](#) should prevent dynamic rollover.
2. [ARM](#) should prevent static rollover.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when [ARM](#) is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

None.

4 DETAILED DESIGN

This chapter outlines the detailed design of Arm_Ay_Mod.

4.1 ALGORITHM DESCRIPTION

4.1.1 RateLimiting

The function limits the rate of change of the input. It provides different rates for falling and rising signals. According to the defined rise and fall rate, the input is limited.

$$\text{output} = \text{LIMIT}(\text{input}, \text{falling_limit}, \text{rising_limit}) \quad (4.1)$$

4.1.2 MaxSwitch

The sign of a reference signal decides, whether to take the minimum or the maximum of two signals.

$$\text{output} = \begin{cases} \text{MAX}(\text{input}_1, \text{input}_2) & ; \text{if reference} \geq 0 \\ \text{MIN}(\text{input}_1, \text{input}_2) & ; \text{if reference} < 0 \end{cases} \quad (4.2)$$

4.1.3 RawDerivative

Calculates the derivative of a signal by using a simple difference equation.

$$y[k] = \frac{1}{T_s} \cdot (x[k] - x[k - 1]) \quad (4.3)$$

where T_s is the sampling time.

4.1.4 FilterPT1MaxOutput

The FilterPT1MaxOutput is a special version of a PT1 filter. It is only applied, when the signal is falling and not when it is rising. Therefore it follows a rising signal but fades slowly to zero by using a PT1 filter. The filter returns the maximum of the input signal and the filtered input signal.

The transfer function of a standard PT1 filter is

$$H(z) = K \cdot \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1}} \quad (4.4)$$

The equation above leads to the difference equation of FilterPT1MaxOutput

$$\begin{aligned} y[k] &= b_0 \cdot x[k] + w[k - 1] \\ y[k] &= \text{MAX}(y[k], x[k]) \\ w[k - 1] &= b_1 \cdot x[k] - a_1 \cdot y[k] \end{aligned} \quad (4.5)$$

4.1.5 FilterPT1MaxSwitch

The function filters the input signal and uses a reference signal to determine whether to return the maximum or the minimum of the input and the filtered signal.

4.1.6 HPFilterAyMod

This high pass filter uses the modified lateral acceleration [AyMod](#) to limit the input.

4.1.7 ResetFilterPT1States

Reset the first order canonic filter state to a value such that the filter can correctly filter signals with the new coefficient for a infinite amount time.

5 PROVIDED TASKS

None.

6 PROVIDED SERVICES

- [RateLimiting](#)
- [RawDerivative](#)
- [FilterPT1MaxOutput](#)
- [FilterPT1MaxSwitch](#)
- [HPFilterAyMod](#)
- [ResetFilterPT1States](#)

6.0.1 Private Services

- [MaxSwitch](#)

7 CONFIGURATION OPTIONS

None.

8 CONFIGURATION PROCEDURES

None.

9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

None.

11 APPENDIX